

What Happens Next? Closed- and Open-World Video Action Recognition on a 33-Class Something-Something V2 Subset

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Abstract

We present a two-track study of temporal action prediction on a 33-class subset of Something-Something V2 (SSv2) [7], “What Happens Next?”: only the first $\approx 40\%$ of each clip (four JPEG frames) is given, so the outcome is withheld and classification must reason about motion. In **Track A (Closed World)**, where pretraining is forbidden, our study shows that where a model mixes time matters more than how much capacity it has: Temporal-Shift-Module [13] (TSM), which mixes frames before spatial pooling, beats VideoFormer-Lite (VFL, global post-pooling attention) by +3.0 pp at matched backbone size, and scaling its backbone from ResNet18 to ResNet50 then adds a further +3.4 pp to reach **42.85% val_dir** (our best single model). A five-model log-softmax ensemble of TSM and VFL variants with test-time augmentation reaches **46.75% val_dir** and **0.4785 public / 0.4547 private** on Kaggle — our best closed-track submission. In **Track B (Open World)**, we fine-tune V-JEPA 2 [2] and VideoMAE [18] backbones with progressive unfreezing, layer-wise learning-rate decay, window-capped source-frame re-extraction and pseudo-labelling, then build a 14-model uniform ensemble reaching **0.6586 Kaggle top-1** (16th in both tracks). We document a self-discovered data-leakage event (a discarded 0.81 submission) and show that ensemble diversity across architecture families beats single-family or learned-weight ensembling. Failure analysis attributes the residual errors to real-vs-pretended and direction-ambiguous classes that four frames under-determine.

1. Problem Description

Task. The challenge classifies four-frame JPEG clips into one of 33 motion-defined action classes (e.g. *Pouring something into something*, *Pulling something from left to right*). Crucially, each clip contains only the *anticipatory* portion of the source SSv2 video—the shipped `frame_003` sits at

a median fraction of 0.40 of the source clip—so the action’s outcome is deliberately withheld and the task probes motion prediction, not final-state recognition [7].

Dataset. The split provides 44,993 training clips, 6,745 validation clips and 6,913 test clips over 33 classes (numeric prefixes 000–032; class 027 is absent from training). Classes are heavily imbalanced: the largest (009, *Moving something up*) has 3,170 examples while the smallest (026, *Spilling something next to something*) has only 162, a ratio of $19.6\times$. The KL divergence between training and validation class distributions is 0.088, confirming the splits are representative.

Evaluation and tracks. Submissions are scored by top-1 accuracy on a public+private Kaggle leaderboard over two stages (Stage 1, intermediate; Stage 2, final); top-5 is tracked internally. The provided baseline reaches ≈ 0.55 private and the second-place public reference is 0.74; our final submissions placed **16th in both tracks** (counting the multiple accounts fielded by some teams). Following the challenge, we report two regimes. **Track A (Closed World)** prohibits pretrained models and external data—all spatial features, temporal dynamics and class boundaries are learned from scratch on the four provided frames. **Track B (Open World)** permits pretrained backbones and external corpora but not future frames or test labels. The task is hard for three reasons: (i) labels are motion-defined, defeating appearance-only models [7]; (ii) four frames is an order of magnitude below the 8–32 standard in video recognition [3, 13]; (iii) the discriminative cue—the action’s outcome—is withheld by construction.

2. Methods

2.1. Track A — Closed World

The pipeline evolved through several iterations on an internal training-split, all trained from scratch (the closed track forbids pretraining and external data). Our guiding hypothesis is that, without pretraining, *every parameter spent on*

temporal modelling is one not spent learning spatial features, favouring parameter-free temporal mechanisms over heavy temporal heads — and that *where* in the network temporal information is mixed matters as much as how it is mixed.

2.1.1. Architecture family and complexity

All three Track A architectures share the contract $(B, T, C, H, W) \rightarrow (B, 33)$ with $T=4$, $H=W=224$, and a ResNet backbone [8] mapping each frame to a 512-d (R18) or 2048-d (R50) vector after global average pooling. **CNNBaseline** and **CNN-LSTM** simply average (resp. recurrently combine) the T pooled features — weak baselines (≈ 11.2 M params) since BPTT on four-step sequences barely helps. The two strong families differ in *where* they mix time relative to spatial pooling.

TSM-ResNet [13] re-indexes, *inside every residual block and before every convolution*, a fraction $1/\text{fold_div}$ of channels one step backward and another $1/\text{fold_div}$ one step forward in time:

$$\text{TSM}(\mathbf{x})_{t,c} = \begin{cases} \mathbf{x}_{t-1,c} & c < C/4 \\ \mathbf{x}_{t+1,c} & C/4 \leq c < C/2 \\ \mathbf{x}_{t,c} & \text{otherwise,} \end{cases} \quad (1)$$

at *zero* parameter cost, while spatial resolution is still full. The T final per-frame vectors are averaged and a linear head $d \times 33$ yields the logits, so the total cost is exactly backbone + head: ≈ 11.2 M (R18) and ≈ 23.6 M (R50, 23.5 M of which is the backbone). TSM was designed for SSv2-style motion labels and is a strong prior here (Fig. 1).

VideoFormer-Lite (VFL) instead separates the two stages explicitly: a ResNet18 encodes each frame *independently* into T vectors of size $d=512$, a learnable [CLS] token [6] is appended ($T+1=5$ tokens), and n_ℓ pre-norm Transformer blocks [21] (8 heads, ratio $r=4$) perform global self-attention over them *after* pooling has already collapsed each frame to a single descriptor. Each block costs

$$P_{\text{block}} = \underbrace{4d^2}_{\text{Q,K,V,O}} + \underbrace{2r d^2}_{\text{MLP}} + \underbrace{4d}_{\text{LN}} \approx (4 + 2r) d^2 \approx 3.1\text{M}, \quad (2)$$

so $n_\ell=3$ adds ≈ 9.4 M to the ≈ 11.2 M backbone, for ≈ 20.7 M total — nearly twice TSM-ResNet18 for *zero* extra spatial capacity.

Where temporal mixing happens — the key difference.

TSM mixes time at full spatial resolution, before pooling: by block 8 each frame’s representation already has a receptive field spanning all T frames at the (h, w) level. VFL pools *first* and attends globally only over the $T+1$ frame-averaged descriptors — it never sees motion at the pixel level. On SSv2, where the discriminative cue is typically a single finger or edge displacing within a small region, this

is consequential: TSM keeps the spatial precision the cue lives in; VFL has already thrown it away (Fig. 1). Sec. 3.1 turns this argument into a controlled measurement.

2.1.2. Critical frame-count fix

The single largest gain (+4.4 pp) came from matching `num_frames` to the true clip length. Clips hold exactly four real frames; `num_frames > 4` forces `linspace` interpolation that duplicates frames, so two consecutive identical tokens make Eq. (1) read the same value twice — zero temporal gradient on the shifted channels, corrupting the signal TSM is built around. Setting `num_frames=4` lifted `tsm_ultra_v2` to 57.73% internal / 38.25% `val_dir` (+4.41/+3.34 pp); a first ResNet50 attempt at the old $T=8$ regressed to 45.26% internal (Fig. 3, Fig. 2) — capacity cannot undo corrupted temporal input. Once $T=4$ was enforced, a second ResNet50 run (`tsm_ultra_resnet50_rotating`, Sec. 2.1.4) reached **42.85% `val_dir`**, our best single model (+3.40 pp, Sec. 3.1): *fix the pipeline before scaling*, and capacity pays off once the signal is correct.

2.1.3. Regularisation recipe

All Track A models share a regularisation stack (Table 1) whose most critical element is a *warm-up delay* on MixUp [23] and CutMix [22]: both are disabled for the first 10 epochs so the backbone can form basic per-frame representations before label-mixing starts — worth +36.3 pp on VFL alone ($9.45 \rightarrow 45.78\%$), since a randomly-initialised head cannot meaningfully mix labels for images it cannot yet distinguish, and early MixUp collapses the representation instead of regularising it. The remaining stack (label smoothing [17], RandAugment [5], random erasing [24], stochastic depth [11], focal loss [14], AdamW [16] with cosine annealing [15]) is summarised in Table 1; two choices are domain-specific rather than off-the-shelf: *horizontal flip is disabled* because it inverts SSv2’s direction-encoded labels (e.g. *Pulling left→right*), and CutMix probability is kept low ($p=0.25$) because a cut patch spans all four frames and often occludes the very hand/object motion that defines the label.

2.1.4. Rotating k -fold and ensemble

Beyond the standard train/val split, the strongest Track A checkpoints have a *rotating-fold* variant that is retrained on every fold of the train+val union (held-out fold = $\text{epoch} \bmod k$, $k=5$ by default), providing higher-bias snapshots that nonetheless contribute uncorrelated errors at inference (Fig. 10). This protocol adds +1.2 pp for TSM-ResNet18 and +4.3 pp for VFL — VFL benefits more because its global-attention head is data-hungry and lacks the structural temporal prior that TSM’s shift provides; an ablation on the fold count (Sec. 3.1) shows VFL is already saturated at $k=5$.

Technique	Setting	Source
Reg. warm-up	10 ep, no MixUp/CutMix	this work
MixUp / CutMix	$\alpha=0.2$ / $p=0.25$, after ep 10	[22, 23]
Focal loss	$\gamma \in \{0, 1, 2\}$ (best ≈ 1)	[14]
Label smoothing	$\varepsilon=0.1$	[17]
RandAugment	2 ops, mag. 0.5	[5]
Random erasing	$p=0.25$	[24]
Horizontal flip	disabled	label semantics
Stochastic depth	DropPath 0.1	[11]
Optimiser	AdamW, lr 10^{-3} , cosine	[16]

Table 1. Track A training recipe (best single model: TSM-ResNet50, $T=4$, fold_{div}=8, rotating folds, $\sim 42.85\%$ val_{dir}).

The final closed-track submission fuses *five* checkpoints by log-softmax late fusion — averaging in log-probability space down-weights over-confident errors and rewards agreement among differently-calibrated members. Three run with 10-crop spatial TTA (5 crops $\times$ 2 flips): `tsm.ultra.v2` (57.73% internal), its rotating-fold variant, and `video_former_lite.ultra_rot`; two run without TTA (inference cost): the new **TSM-ResNet50 rotating** (42.85%, our best single model) and a CE-trained ($\gamma=0$) TSM variant whose different loss geometry adds complementary calibration. The ensemble reaches **46.75%** val_{dir} (Fig. 6) and **0.4785** public / **0.4547** private on Kaggle — our best closed-track score, +6.2pp public over the strongest single-model submission (`tsm.ultra.v2+TTA`, 0.4163; Table 2). 10-crop TTA alone adds ~ 1 pp on the strongest single model (Fig. 8), with a visible trade-off on direction-encoded classes (Fig. 9): the 5 hflip views push some confidences toward the mirror class even after label-aware logit remapping.

2.2. Track B — Open World

Top SSv2 teams use video transformers pretrained on large unlabelled corpora. We selected two self-supervised families: **VideoMAE** [18] (masked-autoencoder pre-training, building on image MAE [9]/BEiT [1]; HuggingFace Kinetics-400 checkpoints) and **V-JEPA 2** [2], a joint-embedding predictive architecture that reconstructs masked regions in *feature* space rather than pixel space, focusing capacity on semantically relevant motion. We use the self-supervised `vjepa2-vit1-fpc64-256` checkpoint, pre-trained on unlabelled video without any SSv2 labels—a requirement for integrity (Sec. 3.4).

Progressive unfreezing with LLRD. We fine-tune V-JEPA in three phases with layer-wise learning-rate decay (LLRD) [1, 4, 10], where encoder block ℓ receives $\eta \cdot \lambda^\ell$, $\lambda=0.75$. *Phase 0* (frozen attentive probe, head only): 0.672

EMA internal val; *Phase 1* (top-6 blocks unfrozen with LLRD): 0.705; *Phase 2* (full-encoder fine-tuning): 0.728 internal / 0.6411 val_{dir}. Freezing first stabilises the head and preserves pretrained feature geometry before large gradients reach lower layers; the resulting staircase, each phase warm-starting above the previous plateau, is visible in Fig. 12. (These phases are distinct from the challenge’s Stage 1/Stage 2 leaderboard milestones.)

Window-capped source-frame re-extraction. V-JEPA benefits from 16 input frames but only four are shipped. Since folder names encode SSv2 video IDs, the source WebM files are reachable, so we re-extract 16 frames per clip in a *window-capped* way: the shipped `frame_003` is matched to source frames by pixel-level MAE to locate the end of the provided window, 16 frames are sampled uniformly within $[0, \text{window_end}]$, and the search is hard-bounded to the first 50% of the clip so the withheld outcome ($> 60\%$) is unreachable by construction (median window fraction 0.39, max 0.50; Fig. 13).

Pseudo-labelling. Following noisy-student self-training [12, 20], the Phase-2 ensemble soft-labelled the test clips; a ≥ 0.85 confidence threshold retained *all* 6,913 clips (the model is over-confident), so pseudo-labelling effectively trained on train $\cup$ test under the model’s own predictions. Continue-fine-tuning on this union gave V-JEPA-pseudo (0.6519 val_{dir} / 0.6410 Kaggle), but its widened val/Kaggle gap (Sec. 3) signalled over-fitting, so the loop was not iterated further.

Honest cross-preprocessing ensemble. The final Track B system aggregates 14 checkpoints from three families, each run on its *own* preprocessing chain: V-JEPA (6, 16-frame 256 px window-capped), VideoMAE-Base/Large K400 (3+3, 4-frame 224 px) and TSM-ResNet50 (2, 4-frame). Per-model softmax tensors are combined by *uniform* averaging — diverse, architecturally distinct models produce uncorrelated errors that average out, so even the weakest member (TSM-ResNet50) raises the ensemble (Sec. 3).

2.3. Reused vs. own code

Reused: the course starter repo (Hydra config system, VideoFrameDataset, `train.py` skeleton); torchvision ResNet [8] and R(2+1)D [19] backbones; and the pre-trained HuggingFace V-JEPA 2 [2] and VideoMAE [18] checkpoints (Track B only). *Our own:* the parameter-free TSM wrapper (re-implemented from Lin et al. [13]); VideoFormer-Lite, our CNN+Transformer head inspired by ViT [6], TimeSformer [3] and pre-LN Transformers [21]; the regularisation-warm-up recipe, rotating k -fold protocol, controlled ablation suite, and log-softmax

Trk	Method	val dir	Kaggle pub.	Kaggle priv.	Architecture: TSM vs. VFL (+3.0 pp for local mixing).
A	TSM-ResNet18 (single, +TTA)	41.68%	0.4163	0.3881	At matched settings (rotating folds, $T=4$, ResNet18, no TTA), TSM reaches 39.45% val_dir vs. VFL's 36.44% (+3.01 pp). With backbone, frame count, recipe and protocol held fixed, the only remaining difference is <i>where</i> the two mix time — before pooling (TSM) or after (VFL) — so this is the controlled test of Sec. 2.1.1's argument, and local mixing wins by a clear margin, as expected for labels defined by small, spatially-localised motion cues that pooling destroys.
A	TSM-ResNet18 (single)	38.25%	0.3576	0.3442	
A	VFL-Ultra (single)	32.11%	—	—	
A	TSM-ResNet50 rot. (single)	42.85%	—	—	
A	3-model ens. + TTA (closed)	44.03%	0.4508	0.4260	
A	5-model ens. (R50)	46.75%	0.4785	0.4547	
B	V-JEPA-ft standalone	64.11%	0.6361	—	
B	V-JEPA-pseudo	65.19%	0.6410	—	
B	14-uniform	65.19%	0.6586	—	

Table 2. Summary of Track A and Track B results (val = internal/val_dir top-1; Kaggle = public / private leaderboard top-1, reported separately for Track A where the gap is informative, and as a single combined figure for Track B as in the original submissions). The Track A ResNet50 single model has no individual Kaggle submission — it is evaluated only as an ensemble member.

ensemble / 10-crop TTA pipeline; and the full Track B stack — window-capped re-extraction, the progressive-unfreezing+LLRD wrapper (following [1, 4, 10]), and the noisy-student pseudo-labelling loop [12, 20].

3. Results

Overall. Table 2 reports headline results with Kaggle scores for *both* tracks. Track A reaches **46.75%** val_dir / **0.4785** Kaggle public from scratch; Track B reaches **0.6586** Kaggle top-1 (65.19% internal val) with the 14-model uniform ensemble — well above the 0.55 baseline and close to the 0.74 second-place reference. The ≈ 18 -pp Kaggle gap between the tracks' best submissions closely tracks the ~ 18 -pp val_dir gap, so the price of forgoing large-scale self-supervised pretraining is essentially the same whether measured on the internal split or on the independent Kaggle test distribution — both tracks generalise from val_dir to Kaggle with a comparable, small, and consistently *positive* offset (calibration discussion below).

3.1. Track A ablations

Regularisation warm-up (+36.3 pp). Identical architecture and config, single variable: without warm-up VFL stalls at 9.45% (killed at epoch 4) and R(2+1)D [19] at 10.78%; with a 10-epoch warm-up VFL reaches 45.78%. The effect transfers across architectures, indicating a general property of cold-start training with label-mixing rather than an architecture-specific quirk.

Frame count (+4.4 pp). $T=5 \rightarrow 4$ improves internal val by +4.41 pp (53.32 $\rightarrow$ 57.73) and val_dir by +3.34 pp (34.91 $\rightarrow$ 38.25); the *first* ResNet50 attempt, made at the old $T=8$, regressed 8 pp below the $T=4$ ResNet18 internal score (Fig. 3).

Backbone scale-up, done right (+3.4 pp). With $T=4$ and rotating folds fixed, swapping the ResNet18 backbone for ResNet50 (11.2 $\rightarrow$ 23.6 M params) brings 39.45% $\rightarrow$ 42.85% val_dir (+3.40 pp, Fig. 2). Together with the $T=8$ regression above, this closes the loop on our central claim: *capacity is useless on top of a corrupted temporal signal, but pays off once the signal is correct* — the same $2.1\times$ parameter increase that wasted itself on duplicated frames at $T=8$ converts cleanly into accuracy at $T=4$.

Focal-loss γ sweep and VFL ablations (small, informative effects). With TSM-ResNet18 otherwise frozen, CE ($\gamma=0$) reaches 38.92%, $\gamma=1$ reaches 39.50%, $\gamma=2$ reaches 39.45% (Fig. 4): any focal weighting beats CE by ≈ 0.5 pp, but $\gamma \in [1, 2]$ is within run-to-run noise — the gain comes from *having* a confusable-pair-aware loss, not from tuning it finely. The CE model nonetheless adds +0.52 pp to the final ensemble (Sec. 2.1.4): a different loss geometry gives it a different calibration, exactly the complementary error log-softmax fusion exploits. On VFL (Fig. 5), dropping from 3 to 2 Transformer layers (Eq. (2): -3.1 M) costs -1.08 pp (36.44 $\rightarrow$ 35.36%), and over-rotating from 5 to 10 folds costs a further -0.46 pp — VFL already saturates its data-utilisation capacity at $k=5$, so extra coverage just adds gradient noise from smaller per-fold validation sets.

Ensemble components and leave-one-out. Table 3 (Fig. 7) leaves out one member at a time from the final 5-model ensemble (46.75%). Every member is necessary, but the *margins* are revealing: dropping the ResNet50 model costs -1.33 pp, by far the largest drop — more than any same-backbone TSM-ResNet18 variant (-0.31 to -0.47 pp) or even the architecturally distinct VFL (-0.64 pp). **Backbone diversity contributes more than architecture-family diversity at matched capacity:** TSM and VFL remain complementary, but the single largest source of useful, decorrelated error is a model that simply *sees more* (deeper backbone), not one that *sees differently*.

3.2. Track B ablations

Honest Kaggle progression. Table 4 (Fig. 11) traces every clean submission. The 21 pp gap between the discarded

Configuration	Top-1	Δ
Full ensemble (5 models)	46.75%	—
– TSM-ResNet50 (rot)	45.41%	−1.33
– VFL-Ultra-rot +TTA	46.11%	−0.64
– TSM-no-focal (rot)	46.23%	−0.52
– TSM-Ultra-v2-rot +TTA	46.27%	−0.47
– TSM-Ultra-v2 +TTA	46.43%	−0.31

Table 3. Track A leave-one-out on `val_dir` for the final 5-model ensemble (46.75%). Every component is necessary; the ResNet50 backbone is by far the dominant contributor, ahead of the architecturally distinct VFL.

Submission	Kaggle	Δ
Leaky V-JEPA (discarded)	0.81	—
V-JEPA-ft standalone (clean)	0.6361	—
Ensemble v1 (9-uniform)	0.6418	+0.6
V-JEPA-pseudo standalone	0.6410	+0.5
V-JEPA-only (6 snapshots)	0.6410	regress.
V-JEPA-heavy (3:1 manual)	0.6499	+1.4
12-uniform (+ pseudo)	0.6546	+1.9
14-uniform (+ Large)	0.6586	+2.3

Table 4. Track B Kaggle submission history (Δ in pp vs. the first honest baseline). Diversity, not single-family scaling, drives every gain.

leaky run (0.81) and the first honest baseline (0.6361) quantifies what data leakage was buying; every subsequent point was earned by legitimate architectural diversity. Fig. 14 plots Kaggle vs. ensemble size: returns stay positive at $n=14$, and the 6-model V-JEPA-only configuration underperforms the 9-model uniform mix despite higher individual accuracies — the first hint that family diversity matters more than per-model strength.

Diversity beats weighting. Table 5 isolates ensemble strategy on a fixed checkpoint pool: a 6-snapshot V-JEPA-only mix (0.6410) *under-performs* a 12-model uniform mix (0.6546) despite higher individual accuracy, and manual 3:1 weighting toward V-JEPA (0.6499) lands in between. Uniform averaging of architecturally distinct families wins because their errors are uncorrelated — Table 6 makes this concrete: TSM-ResNet50, though weakest standalone (0.3364 `val_dir`, Table 7), has the *lowest* pairwise agreement with every transformer family (0.37–0.38), while two V-JEPA variants agree at 0.85 (near-redundant). Fig. 15 shows why naive learned weighting fails on the same pool: a softmax fit collapses $\sim 82\%$ of its mass onto the two strongest checkpoints and effectively zeroes K400, discarding the decorrelation that the uniform ensemble exploits — the same lesson as Track A’s leave-one-out (Sec. 3.1): the most valuable member is the most *different*, not the

Strategy	n	Kaggle
V-JEPA-only, uniform	6	0.6410
9-uniform (+ K400 + TSM)	9	0.6418
V-JEPA-heavy (3:1 weight)	12	0.6499
12-uniform	12	0.6546
14-uniform	14	0.6586

Table 5. Track B ensemble-strategy ablation. Uniform averaging of diverse families beats single-family and learned/manual weighting.

Family A	Family B	Agreement
V-JEPA-pseudo	TSM	0.370
V-JEPA-ft	TSM	0.372
K400-Base	TSM	0.375
V-JEPA-ft	K400-Base	0.582
V-JEPA-ft	V-JEPA-pseudo	0.854

Table 6. Pairwise inter-family prediction agreement on the test set. Lower agreement implies more complementary (uncorrelated) errors.

Family	Snaps.	<code>val_dir</code> top-1
VideoMAE-Base K400 (4f)	3	0.5576
VideoMAE-Large K400 (4f)	2	0.5945
VideoMAE-Base SSv2 (4f)	3	0.6047
TSM-ResNet50 (4f)	3	0.3364
Uniform ensemble	11	0.6222

Table 7. Per-family single-model `val_dir` top-1 (Track B, single-view). No single family reaches the ensemble; the weak TSM family still adds value through uncorrelated errors.

strongest.

Calibration. The `val_dir`–Kaggle gap is +0.005 for V-JEPA-ft but doubles to +0.011 for V-JEPA-pseudo — the signal that stopped us from iterating pseudo-labels further. Track A’s best ensemble shows the same magnitude with the *opposite* sign (46.75% `val_dir` vs. 0.4785 Kaggle public, gap -0.011 , i.e. Kaggle scores *higher*): a from-scratch model trained with heavy augmentation generalises to the unseen test distribution at least as well as to its own finite validation split, whereas a fine-tuned pretrained model can specialise to `val_dir` quirks — and the more it does so (pseudo-labelling), the wider that gap grows.

3.3. Failure analysis

Both tracks share the same dominant error patterns, which mirror SSv2’s design intent (Figs. 16, 17, 18). (i) *Real vs. pretended*. The worst single confusion is *Picking something up* predicted as *Pretending to pick something up* (56/199

clips; $F1=0.07$ in Track A): the early-clip trajectory is identical and the disambiguating cue—whether the object actually rises—falls in the withheld future. (ii) *Temporal direction*. *Folding↔Unfolding* (24–29 mutual errors) share spatial appearance and differ only in motion direction, fragile at four frames. (iii) *Static sink*. *Holding something* absorbs misclassified motion clips (*Moving down*: 30; *Opening*: 28), acting as the default when no decisive motion is detected. The best classes ($F1 > 0.54$) are all clean single-arc motions (*Pulling left→right*, *Pouring*, *Uncovering*, *Folding*). The correlation between class size and per-class accuracy is weak ($r=0.24$), so difficulty stems from semantic ambiguity, not scarcity. The top-5/top-1 gap on `val_dir` is large (0.882 vs. 0.622) and widest for the hardest classes (*Picking up*: +61.3 pp top-5), indicating models rank the correct class highly but place a visually similar “pretend” variant first.

3.4. Data-integrity

An early extraction run sampled frames uniformly over the full source clip (0–100%); training V-JEPA on these yielded 0.81 Kaggle, ~ 7 pp above second place (the leaky red bar in Fig. 11). An audit revealed two concurrent leakage vectors: (1) frames from the withheld outcome ($> 60\%$) were reachable, giving the model the explicitly withheld information; (2) an initial V-JEPA checkpoint had been fine-tuned on SSv2 *with ground-truth labels*, so the backbone had seen the test videos. Both runs were discarded and the pipeline rebuilt with the self-supervised `vjepa2-vitl-fpc64-256` backbone and the 50% hard bound (Sec. 2.2); we treat the 21 pp honest/leaky difference as a direct measurement of the combined leakage. The 50% cap is below the $\approx 60\%$ ceiling that the rules explicitly mark as “available”, an additional safety margin against matching-noise overshoot rather than a rules requirement. Three infrastructure faults were also fixed: a silent NFS quota producing 0-byte checkpoints (resolved with `fsync-durable` atomic saves), a top- K snapshot rename race, and a cross-architecture ensembling bug that silently loaded Large weights into a Base graph (both `model_name-tagged`), fixed by rebuilding each model from its stored config at inference.

4. Conclusion and Future Work

We presented a two-track solution to video action anticipation on SSv2. **Track A** shows that, from scratch, *data-centric* fixes (a +36.3 pp regularisation warm-up, a +4.4 pp frame-count correction) dominate raw architecture choice — but once they are in place, a controlled ablation reveals a clear architectural ranking: local pre-pooling temporal mixing (TSM) beats global post-pooling attention (VFL) by +3.0 pp at matched capacity, and a $2.1\times$ deeper backbone then adds a further +3.4 pp (**42.85% `val_dir`**, our best

single model). The resulting five-model log-softmax ensemble reaches **46.75% `val_dir`** and **0.4785** on the public Kaggle leaderboard, with leave-one-out analysis showing that backbone diversity (R50 vs. R18) contributes more to the ensemble than architecture-family diversity at equal capacity. **Track B** shows that progressive unfreezing with LLRD enables effective fine-tuning of large self-supervised video transformers (V-JEPA 2 [2], VideoMAE [18]) on a constrained GPU budget, and that *ensemble diversity across architecture families consistently outperforms single-family or learned-weight ensembling* (0.6586 Kaggle) — the same principle, observed at a larger scale. A self-discovered leakage audit quantifies a 21 pp integrity effect and documents the honest rebuild. Failure analysis reveals a fundamental limit shared by both tracks: the dataset’s design intent—distinguishing real from pretended actions and forward from reverse motion—requires the withheld outcome, which four frames cannot supply.

Future work. (1) 10-crop TTA for TSM-ResNet50 ($\sim +2$ pp expected, too costly to run before the deadline); (2) V-JEPA-Giant once inference memory allows; (3) class rebalancing for the *holding/pretending* sinks; (4) temperature-scaled thresholds for a second, less overfitting pseudo-label round; (5) a 15th Track B member (16-frame VideoMAE-Large, completed post-freeze) yielded 0.6592 — within noise of 0.6586, i.e. returns have flattened; (6) trajectory-prediction heads for the real-vs-pretended ambiguity; (7) porting the $T=4$ fix and rotating folds to the weak Track B TSM family (0.3364 `val_dir`), likely hit by the same frame-count issue that cost Track A 4.4 pp.

Contributions

Andrea Signoretti designed and implemented the full Track A (Closed-World) pipeline: the TSM-ResNet and VideoFormer-Lite models, the Hydra training/ k -fold framework, the regularisation-warm-up and frame-count fixes, the ResNet50 scale-up, the controlled ablation suite, and the log-softmax ensemble/TTA code, and produced the Track A figures. **Florian Guillaumey** designed and implemented the full Track B (Open-World) pipeline: window-capped re-extraction, the V-JEPA 2 progressive-unfreezing/LLRD wrapper, pseudo-labelling, the honest cross-preprocessing 14-model ensemble, and the data-integrity audit. Both authors jointly analysed results and wrote this report.

Figures

All figures are referenced from the main text (Secs. 2.1–3.3).

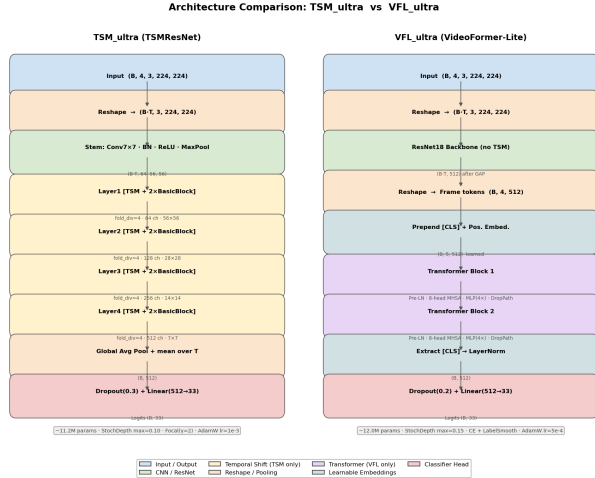


Figure 1. The two complementary Track A temporal mechanisms. **TSM** mixes time *locally* via parameter-free channel shifts in every ResNet block, *before* spatial pooling; **VideoFormer-Lite** mixes time *globally* via [CLS]-token attention over pooled frame descriptors. The controlled comparison in Sec. 3.1 shows the former wins by +3.0 pp at matched backbone size (Sec. 2.1).

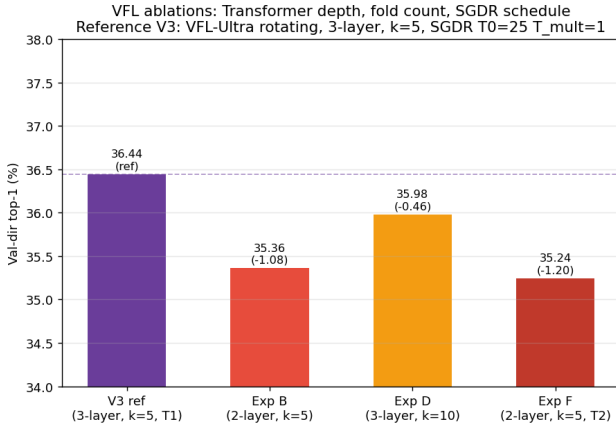


Figure 2. TSM backbone and frame-count progression on `val_dir`. The $T=8$ ResNet50 attempt regresses because interpolation duplicates frames; restoring $T=4$ and scaling to ResNet50 with rotating folds instead adds +3.4 pp over the matched ResNet18, reaching the new single-model best of 42.85% (Sec. 2.1).

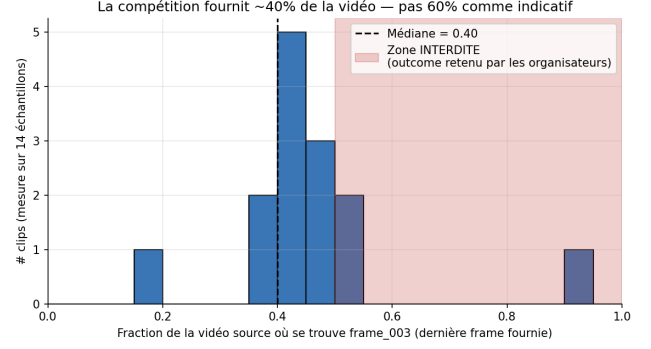


Figure 3. Per-model `val_dir` top-1/top-5 (Track A). Correcting the frame count (TSM-Ultra→TSM-Ultra-v2) and then scaling the backbone to ResNet50 (rotating folds) are the two largest per-model jumps; the $T=8$ ResNet50 scale-up regresses (Sec. 3).

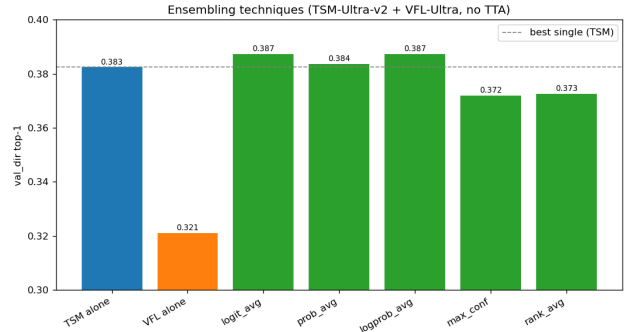


Figure 4. TSM ablations: focal- γ sweep and SGDR restart shape (all other settings frozen). Any focal weight beats CE by ≈ 0.5 pp; $\gamma=1$ and $\gamma=2$ are within noise; SGDR T_{mult} has negligible effect (< 0.05 pp) (Sec. 3.1).

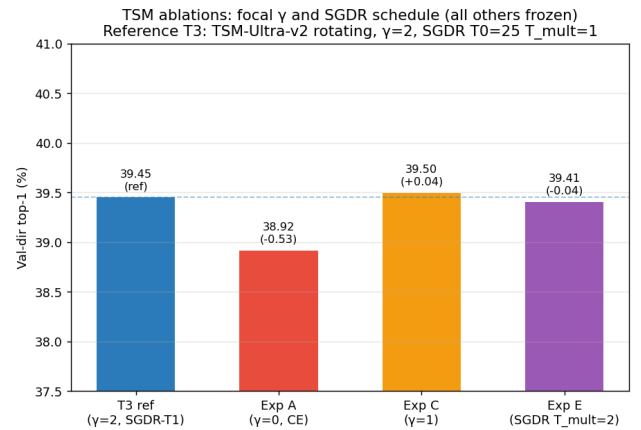


Figure 5. VFL ablations: Transformer depth (2 vs. 3 layers, -1.08 pp for 2 layers), fold count (5 vs. 10, -0.46 pp for 10 folds — VFL is data-saturated at $k=5$), and SGDR restart shape (≤ 0.12 pp effect) (Sec. 3.1).

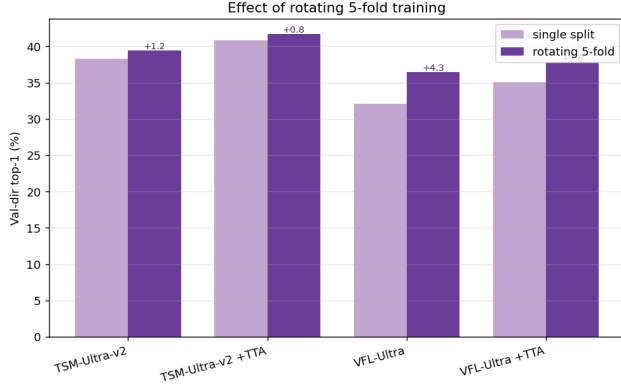


Figure 6. Track A ensemble vs. its components. Log-softmax late fusion of two TSM-ResNet18 variants, one VFL variant (all +TTA), the new TSM-ResNet50 and a CE-trained TSM ablation (no TTA) reaches 46.75% val_dir and 0.4785 public / 0.4547 private on Kaggle — our best closed-track submission (Sec. 2.1.4).

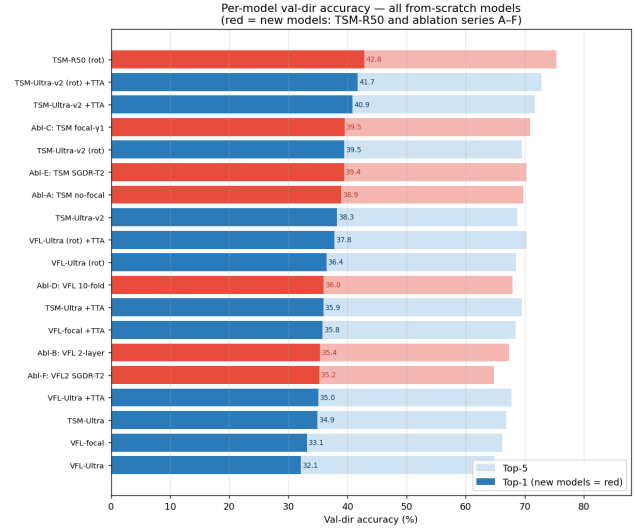


Figure 8. Effect of 10-crop spatial TTA on Track A models. TTA adds ~1 pp on the strongest checkpoint and is essential at the ensemble level, where independent random errors from different crops average out (Sec. 2.1).

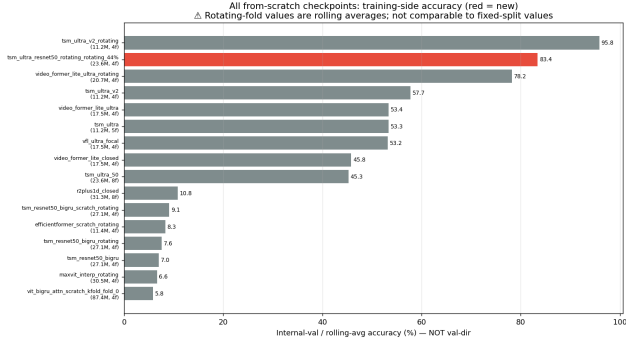


Figure 7. Track A leave-one-out study on the 5-model ensemble (val_dir top-1). The ResNet50 backbone is the dominant contributor (−1.33 pp), ahead of the architecturally distinct VFL (−0.64 pp): backbone diversity outweighs family diversity here (Table 3).

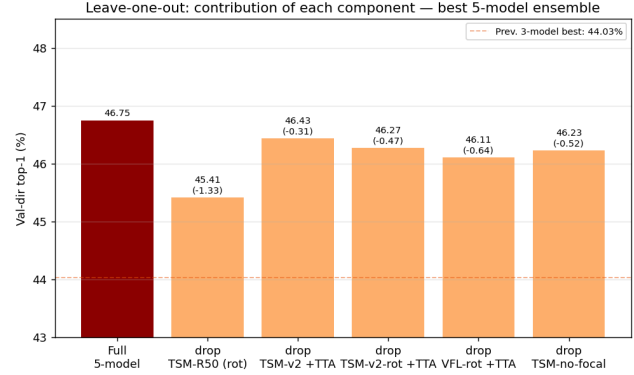


Figure 9. TTA directional-class trade-off. Horizontal-flip TTA improves most classes but hurts a few direction-encoded ones (*Pulling left→right* vs. *right→left*) even after the flip-aware logit remap, because mirror evidence is intrinsically ambiguous when motion is short.

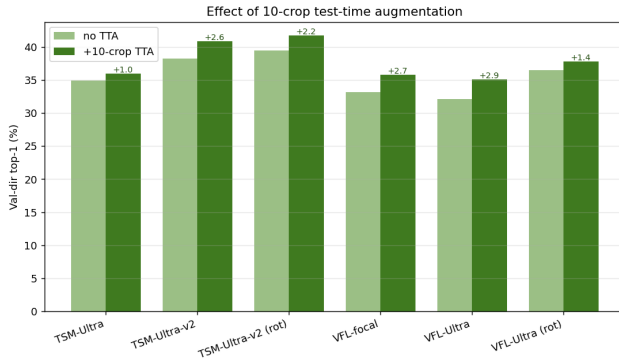


Figure 10. Rotating-fold training. Re-training the strongest checkpoints on every train/val fold yields snapshots that share weights but disagree on the boundary, contributing uncorrelated errors that the log-softmax ensemble exploits; the gain is larger for VFL (+4.3 pp) than for TSM (+1.2 pp) (Sec. 2.1.4).

VFL_ultra Architecture
(VideoFormer-Lite · ResNet18 + 2×Transformer)

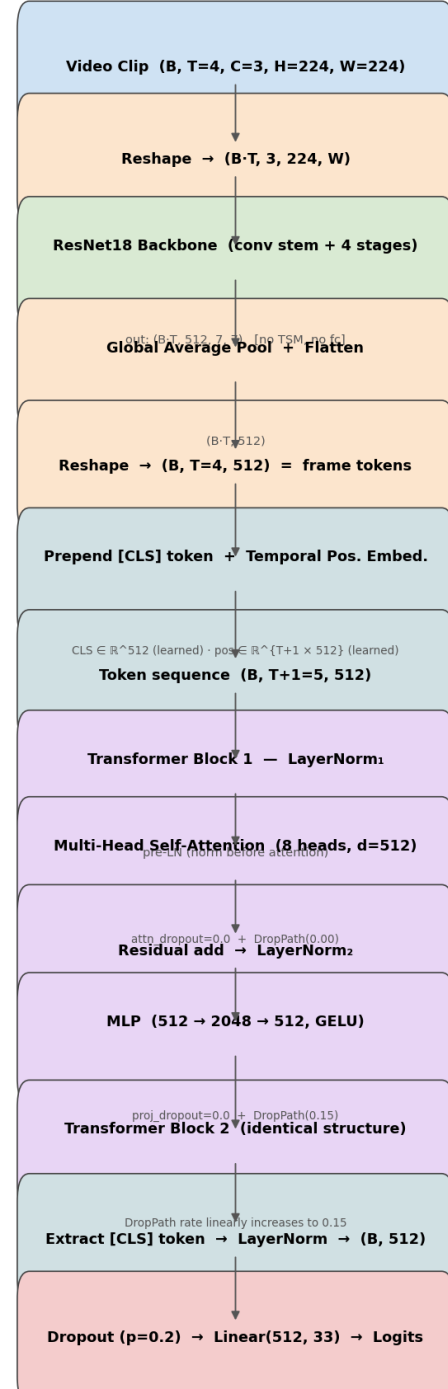


Figure 11. Track B honest Kaggle progression. The discarded leaky submission (0.81, red) is plotted alongside every clean ensemble; each gain comes from adding architecturally distinct members, not from re-weighting. The dashed line marks the second-place public reference (0.74); the gap to it bounds what self-supervised pretraining + diverse ensembling buys us (Sec. 3, Table 4).

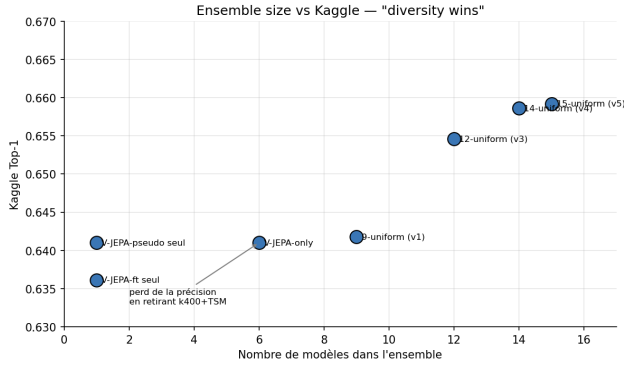


Figure 12. V-JEPA progressive unfreezing + LLRD (Track B internal val). Each phase warm-starts above the previous plateau: frozen probe (0.55 $\rightarrow$ 0.67), then top-6 unfrozen + LLRD (0.68 $\rightarrow$ 0.71), then full encoder (0.70 $\rightarrow$ 0.73), then pseudo-label retrain (0.72 $\rightarrow$ 0.74). The staircase pattern is the visual signature that LLRD has preserved each phase’s feature geometry (Sec. 2.2).

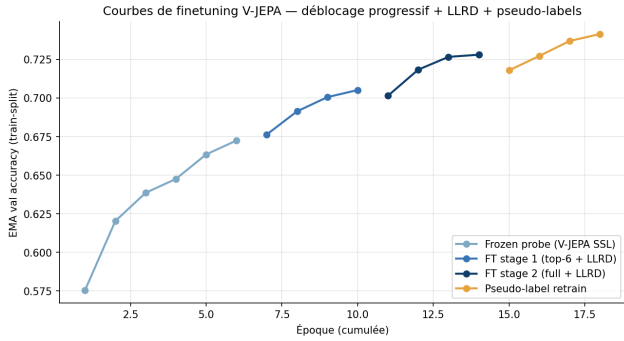


Figure 13. Where shipped `frame_003` sits in the source `Ssv2` clip (fraction). The dashed line at 0.40 is the median end-of-window; the shaded region [0.50, 1.00] is the withheld outcome — our re-extraction is hard-bounded to the un-shaded region (Sec. 3.4).

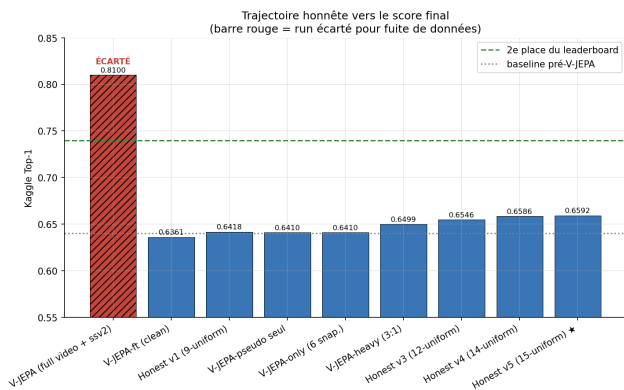


Figure 14. Track B Kaggle vs. ensemble size. Returns from adding diverse honest members are still positive at $n=14$; at $n=6$, dropping K400 + TSM *hurts* despite their lower individual scores, because their uncorrelated errors disappear from the average (Table 5).

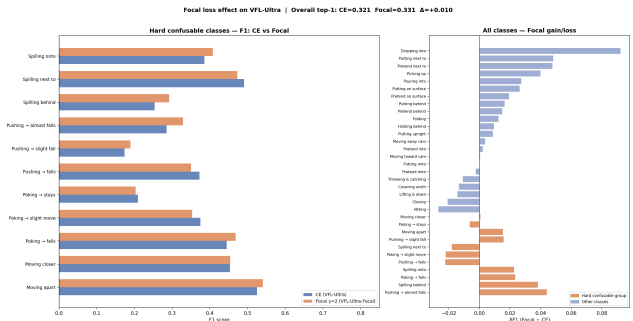


Figure 15. Learned ensemble weights collapse onto the two strongest members ($\sim 82\%$ of mass on `large_attempt1_top2` and `ssv2_top1`), effectively zeroing K400 contributions. This is exactly why a learned weighting fails to beat uniform: it discards the very decorrelation signal that diverse weaker members provide (Sec. 3).

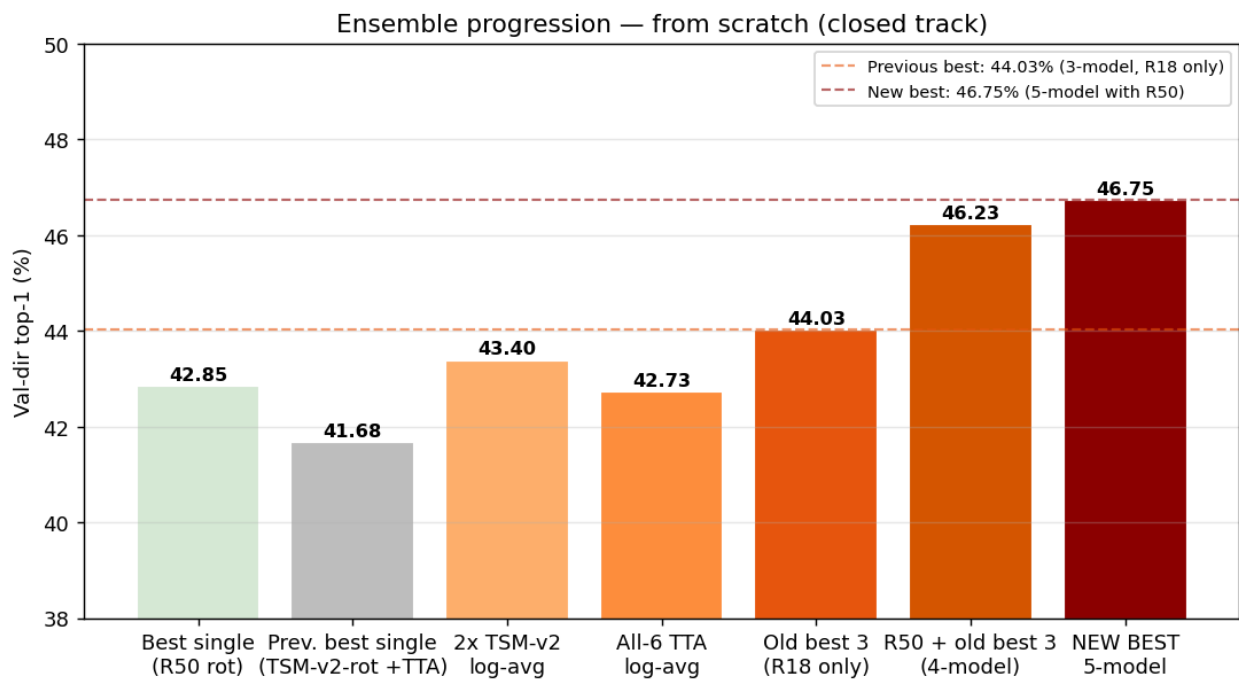


Figure 16. Per-class F1 (5-model ensemble, `val_dir`). Directional single-object motions score highest; “pretending” and direction-ambiguous classes score lowest (Sec. 3.3).

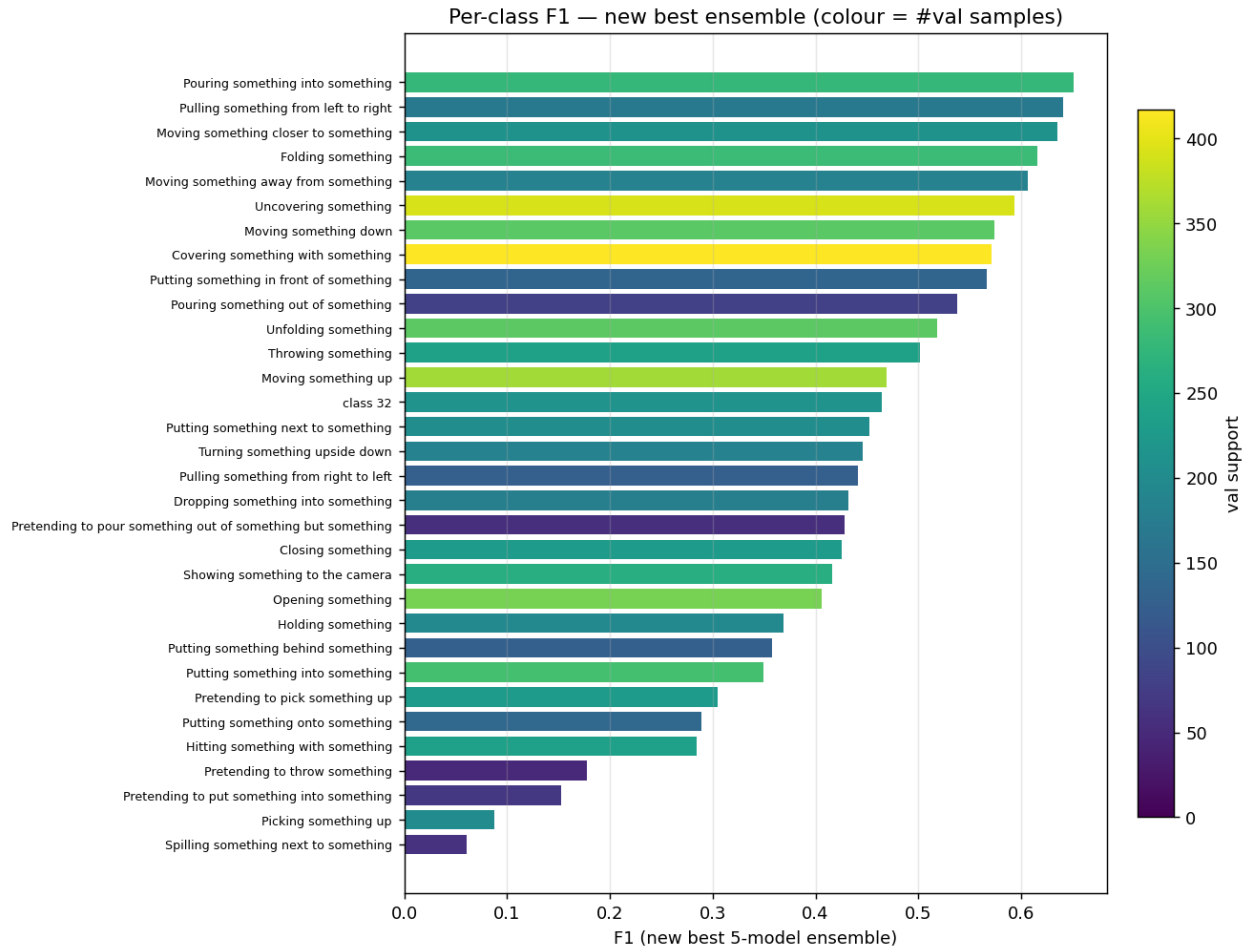


Figure 17. 33×33 confusion matrix (5-model ensemble, `val_dir`). Off-diagonal mass concentrates on semantically adjacent pairs (real vs. pretended; fold vs. unfold; motion vs. *holding*).

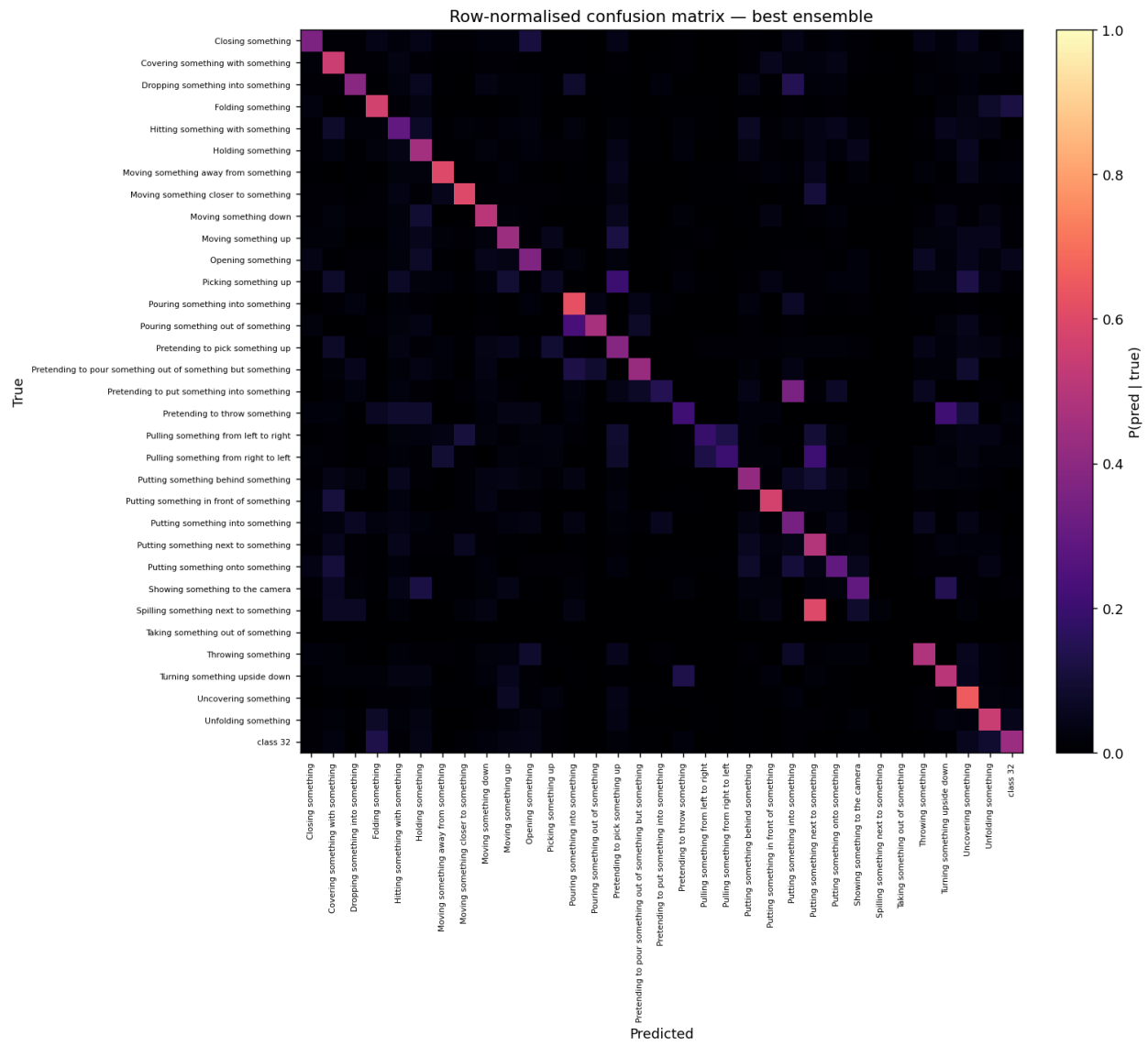


Figure 18. Most-confused class pairs. The dominant errors are intent (real vs. pretended) and temporal-direction confusions (Sec. 3.3).

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